

Underactuated attitude tracking control of tethered spacecraft for deployment and spin-up

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Abstract

This paper investigates the underactuated attitude tracking control problem of tethered spacecraft during tether deploying and spinning. The attitudes are controlled by the torque inputs along two body axes and perturbed by the tether tension torque. The coupling effect between spacecraft attitudes and flexible tethers can be alleviated by simplifying control inputs. The main contribution is the development of an underactuated tracking controller with an adaptive barrier function that inhibits unknown disturbance. To avoid singularity problems with tether and spacecraft during simulation, the dynamic model of a tethered system is derived using quaternion transformations. The underactuated attitude control strategy is designed by kinematic state coupling relations and dynamic backstepping technique, which is given extensive stability demonstration using Lyapunov's theory. An adaptive controller gain with barrier function is employed to inhibit the bounded unknown disturbance with no need for the online estimation signal. Finally, numerical simulations demonstrate the effectiveness and robustness of the proposed underactuated strategy in the case of tether deployment and spin-up, respectively.

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Keywords: Underactuated attitude control; Backstepping control; Barrier function; Tethered spacecraft system; Tether deployment and spin-up

1. Introduction

Tethered spacecraft have been important space vehicles in the last two decades, and play a key role in space missions. It can contribute to a variety of space missions, such as payload delivery, debris removal, space station de-orbiting, etc (Williams (2008); Aslanov and Ledkov (2017)). Generally speaking, the smooth deployment and spin-up are the foundation of space tethered missions. In the past two decades, tethered spacecraft research has been addressed directly to control tether deployment and spin-up with tether tension and nozzle thrusts (Hong and

Varatharajoo (2015b); Mohsenipour (2020); Kang et al., (2021); Li et al. (2020)). The spacecraft attitudes are highly coupled to the tether motion due to the flexibility and elasticity of the tether. The tension torque of tethered spacecraft has a significant effect on attitudes during tether deployment and spin-up, except that the tension torque component in the same direction as the tether has a smaller effect on attitudes (Zhong and Zhu (2019); Chang et al. (2010)). With the aforementioned issues in mind, an increasing number of researchers are concentrating on the tether and 6-DOFs spacecraft coupled motion control, but mostly in full actuated attitude control. The underactuated attitude control technique (Qian et al. (2019)) is a control strategy with fewer control inputs than state variables to make it easier to maintain attitude stability. In view of this, underactuated control can be applied to the tethered spacecraft without the need for torque in the direction

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Nomenclature

A, B	Centroid of sub-spacecraft	λ	Gain of barrier function
a, b	Attachment points of tether on spacecraft	O	Earth's center
α, β	In-plane and out-of-plane angles of orbital plane	$\hat{q}_{or}$	Orbital quaternion
C	Centroid of STS	$\hat{q}_{te}$	Tether quaternion
D_K	Unknown bounded disturbance torques on spacecraft	$\hat{q}_K$	Attitude quaternion of sub-spacecraft
$\hat{e}_K$	Attitude tracking error of spacecraft	r_A, r_B, r_C	Radius vectors of spacecraft and centroid of STS
e_a, e_u	Actuated and underactuated error state signal	r_i	Radius vectors of discrete tether points
$\tilde{e}_d$	The designed auxiliary converge signal	r_{ab}	The vector of tether
$\tilde{e}_a$	Virtual error signal	$\mathbf{r}$	The vector in orbital coordinate system
EA	Axial rigidity of tether	R_K	Rotation matrix from frame K - $x_K y_K z_K$ to frame K_d
F_A, F_B	Control forces on spacecraft	s	Constraint of independent variable of barrier function
f_z	Spin-up thrust	T_{ij}	Tensions of each segmented tether
$\underline{H}_B$	Natural logarithmic barrier function	τ_K	Control torques on spacecraft
i	Orbital inclination	$\sigma_{i,i+1}$	Unit vectors in the direction of the segmented tethers
J_K	Inertia moment of spacecraft	$\bar{u}$	Argument of perigee
$K(\eta)$	Logarithmic adaptive gain	μ	Earth's gravitational constant
k, k_p, k_d	Parameters of tension controller for deployment	ϑ	True anomaly
k_u, k_a	Control gain of kinematic controller	ω_K	Angular velocity of sub-spacecraft
k_η	Control gain of dynamic controller	ω_{Ke}	Angular velocity tracking error of sub-spacecraft
k_{BF}	Parameter of adaptive gain	ω_{ea}	Angular velocity of the actuated axes
$\kappa_1, \kappa_2, \Lambda, \xi$	Virtual variables of kinematic controller	ω_{eu}	Angular velocity of the underactuated axis
$L_{i,i+1}$	Length of each segmented tether	$\underline{\omega}_d$	Desired angular velocity
L_K	Unstretched length of each segmented tether	Ω	Longitude of ascending node
L	Length of tether		Damping coefficient of tether
m_A, m_B	Masses of sub-spacecraft	$\gamma_0, \gamma_1, \gamma_2$	Parameters of auxiliary converge signal
m_i	Mass of each tether element		
M_a	Tethered tension torque on sub-spacecraft		
n	Number of elements for tether		
η	Dynamic tracking error variable		

along the least tension torque component. As a result, we propose an underactuated attitude control technique to reduce the harsh coupling between the tether and spacecraft attitude in this paper.

So far, lots of underactuated tethered control researchers have focused on tension control, or deployment control by only pure tension and gravity gradient torque (Pastor-Rodríguez et al. (2017); Xu et al., (2018)). To deploy the tether, the researchers have designed a tension controller based on the assumption that the tether is a rigid rod (Xu et al., (2018); Ma et al, 2018). On this basis, the underactuated tension controllers with virtual signals have been developed to control tether length and angle simultaneously, where the virtual signals are potential energy or dissipation of the tethered system (Kang and Zhu (2019); Wang and Zhang (2019); Zhang and Huang (2019)). Actually, the underactuated control strategies in the field of robotic arms and inverted pendulums are substituted with energy terms commonly (Dixon et al. (2000)). Chung

et al. (Chung et al. (2008)) have suggested a number of non-linear tethered controllers for underactuated tethered formation systems for convergence of the underactuated states. Subsequently, mathematical models which represent the tether as a flexible rod or discrete point masses connected by the elastic-damping structure were developed (Luo et al. (2019); Shi et al. (2019)). To simplify analysis and facilitate controller design, available publications on the underactuated tethered problem are primarily concerned with in-orbital plane tether motion and 2-DOFs tethered spacecraft motion (Trushlyakov and Yudinsev (2020); Vafamand (2020); Yousefian and Salarieh (2017)). Surprisingly, the main spinning tether research works have ignored the out-of-orbital plane motion (Trushlyakov and Yudinsev (2020); Gou et al., (2018)). This is because the tether model based on rotation matrixes has a singularity problem in out-of-plane angle, which is not conducive to research and simulation. Tether rotation is a full-angle motion. No matter how we choose the rotation sequence,

we cannot avoid the singularity problem. Hence, the quaternion theory is introduced to eliminate the special singularity angle in order that we can research the out-of-plane rotation and spin-up control more easily.

In the past decade, the advancements in the field of underactuated systems have sparked interest in nonlinear control algorithms. Because the actuating relationships of different underactuated objects are various, a universally applicable control method has not been reported that can handle all underactuated issues (Choukchou-Braham et al., (2013); Roy et al., (2021)). For symmetric underactuated mechanical systems, Olfati-Saber has presented a normal underactuated form (Olfati-Saber (2002)). The linear controllability and smooth stability were first derived with an underactuated system by Reyhanoglu et al (Reyhanoglu et al., (1999)). By refining the underactuated control theory, various traditional nonlinear control theories have been applied to specific underactuated systems with specific coupling relations. For instance, a linear sliding mode manifold has been presented for cascaded underactuated systems such as a translational oscillator with a rotational actuator and a quadrotor helicopter Xu and Özgüner (2008). Liu et al (Liu et al. (2019)) employed a radial basis function (RBF) adaptive neural network to build an underactuated tracking control for a two-link manipulator with matched and mismatched disturbances. Huang et al. (Huang et al. (2021)) have also investigated the underactuated autonomous underwater vehicle with a neural network input command filter and prescribed performance theories. In summary, nonlinear underactuated control methods have been developed for various underactuated fields suitable for various coupling relationships. As a result, the underactuated attitude control of flexible tethered spacecraft must be presented urgently. The goal of this paper is to examine the feasibility of underactuated attitude control, deployment and spin-up for tethered spacecraft. The proposed underactuated attitude tracking control approach alleviates the coupling connection of attitude torque on tether motion in this way.

In this paper, we present an underactuated backstepping controller to handle the relationship between spacecraft attitudes and tether motion. We introduce the quaternion transformation instead of the rotation matrix to describe orbit, tether, and spacecraft. It can solve the singularity problem for particular tether angles and attitudes in out-of-orbital motion, so that the designed control is more realistic with no need to ignore out-of-plane motion and 3DOF attitudes (Hong et al., (2015)). In addition, a natural logarithmic barrier function is used to prevent the unknown disturbance without the requirement for an estimated signal. The proposed controller's complexity and computational requirements can be reduced using the adaptive barrier function. When the tether is deployed and spun, the Lyapunov theory is used to demonstrate the stability of the proposed underactuated control strategy.

The remainder of this paper is organized as follows. The hypotheses and dynamic model of the tethered spacecraft

system are presented in Section II. The underactuated attitude tracking controllers of kinematic and dynamics models are proposed with stability analysis in Section III. In Section IV, numerical simulations are carried out to verify the effectiveness of the underactuated control method during tether deploying and spinning. Finally, Section V concludes the paper.

NOTATION: $(\cdot)_K, K = A, B$ is the vector subscript of sub-spacecraft A and B. $\|\cdot\|$ denotes the Euclidean norm of vector. $(\cdot)^\times \in R^{3 \times 3}$ and $diag(\cdot) \in R^{3 \times 3}$ represent the skew-symmetric and diagonal matrices of the vector, respectively. $T_{ij} \in R^{1 \times 3}$ denotes the tether tension vector from mass point i to j , that is, $T_{ij} = -T_{ji}$. “ $\otimes$ ” denotes quaternion multiplication. $r, \hat{r}, \tilde{r}$ denote the vectors with respect to the coordinate systems $O - XYZ, C - x_c y_c z_c$, and $K - x_K y_K z_K$, respectively.

2. Dynamic modeling of tethered spacecraft

2.1. Hypotheses and modelling

Before explaining the tethered spacecraft mathematical model, we define the coordinate system associated with the tethered system. Fig. 1 shows the structure of tethered spacecraft and frames. The geocentric inertial coordinate system is defined as $O - XYZ$, which directs standardly as usual. $C - x_c y_c z_c$ is defined by the orbital coordinate system with the origin at the centroid C of the tethered system, x_c -axis points on the radius vector of centroid C , z_c -axis is perpendicular to the orbital plane, y_c -axis is generated with the right-hand rule. $K - x_K y_K z_K$ are defined by the body-fixed coordinate systems of tethered spacecraft A and B . The axes of body-fixed coordinate systems are the same as the principal axes of inertia of spacecraft. Tether coordinate system $C - x_t y_t z_t$ is defined along with the two attachment points on sub-spacecraft. A 3–2–1 rotation sequence with respect to the tether coordinate system is used to obtain tethered spacecraft attitudes. That is to say, rotate ψ along z_K axis firstly, θ along y_K axis secondly, and ϕ along x_K axis finally. Several reasonable assumptions are given as follows:

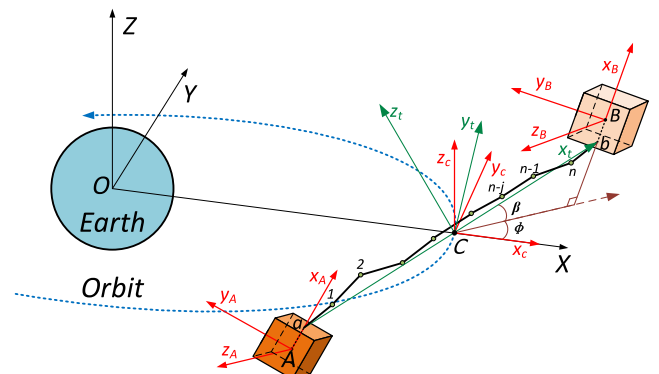


Fig. 1. Schematic of tethered spacecraft and coordinate frames.

- 1) Tether is regarded as a series of discrete point masses linked by incompressible rods with elastic-damping structure;
- 2) Tether is connected on the surface of the spacecraft at points a and b , which are on the principal axes of inertia x_A axis and x_B axis in Fig. 1.

Gravitational force, tethered tension, and torque are the forces and moments operating on tethered spacecraft. Discrete point masses are subject to the gravitational force and tensions on two connected elastic rods. The orbital dynamics model and attitude model of tethered spacecraft are derived using the Newton-Euler kinetic equations based on the above assumptions. The equations of the tethered spacecraft and mass points' motion are given as (1), which is described in the geocentric inertial coordinate system $O - XYZ$.

$$\begin{cases} \ddot{\mathbf{r}}_A = -\mu \mathbf{r}_A / \|\mathbf{r}_A\|^3 + \mathbf{T}_{a,1}/m_A + \mathbf{F}_A/m_A \\ \ddot{\mathbf{r}}_B = -\mu \mathbf{r}_B / \|\mathbf{r}_B\|^3 + \mathbf{T}_{b,n}/m_B + \mathbf{F}_B/m_B \\ \ddot{\mathbf{r}}_1 = -\mu \mathbf{r}_1 / \|\mathbf{r}_1\|^3 + \mathbf{T}_{1,2}/m_1 - \mathbf{T}_{a,1}/m_1 \\ \dots \\ \ddot{\mathbf{r}}_i = -\mu \mathbf{r}_i / \|\mathbf{r}_i\|^3 + \mathbf{T}_{i,i+1}/m_i - \mathbf{T}_{i-1,i}/m_i \\ \dots \\ \ddot{\mathbf{r}}_n = -\mu \mathbf{r}_n / \|\mathbf{r}_n\|^3 - \mathbf{T}_{b,n}/m_n - \mathbf{T}_{n-1,n}/m_n \end{cases} \quad (1)$$

Where, $\mathbf{r}_K, K = A, B$ and $\mathbf{r}_i, i = 1, 2, \dots, n$ are the radius vectors of spacecraft and mass-points. μ denotes the Earth's gravitational constant. $m_K, K = A, B$ and $m_i, i = 1, 2, \dots, n$ denotes the mass of spacecraft and tether. The Kevin-Hoyt model is adopted to obtain the specific forms of segmented tether tension in (2).

$$\mathbf{T}_{i,i+1} = EA(L_{i,i+1} - L_k + Y\dot{L}_{i,i+1})\boldsymbol{\sigma}_{i,i+1}/L_{i,i+1}, \quad (2)$$

$i = a, 1, 2, \dots, n, b$

Where, $\mathbf{T}_{i,i+1}, i = 1, 2, \dots, n-1$ represent the tensions of segmented tether between point i and $i+1$. EA , represent the axial rigidity and damping coefficient respectively. $\boldsymbol{\sigma}_{i,i+1}$ are the unit vectors in the direction of the segmented tethers.

The orbital motions are determined by the classical orbital elements. The orbital coordinate system $C - x_c y_c z_c$ is obtained by a 3–1–3 rotational sequence with respect to the inertial coordinate system $O - XYZ$. That is to say, the orbital quaternion is calculated by the following (3),

$$\hat{\mathbf{q}}_{or} = \begin{bmatrix} \cos(\bar{\Omega}/2) \cos(\bar{i}/2) \cos((\bar{u}+\vartheta)/2) - \sin(\bar{\Omega}/2) \cos(\bar{i}/2) \sin((\bar{u}+\vartheta)/2) \\ \cos(\bar{\Omega}/2) \sin(\bar{i}/2) \cos((\bar{u}+\vartheta)/2) + \sin(\bar{\Omega}/2) \sin(\bar{i}/2) \sin((\bar{u}+\vartheta)/2) \\ \cos(\bar{\Omega}/2) \sin(\bar{i}/2) \sin((\bar{u}+\vartheta)/2) - \sin(\bar{\Omega}/2) \sin(\bar{i}/2) \cos((\bar{u}+\vartheta)/2) \\ \cos(\bar{\Omega}/2) \cos(\bar{i}/2) \sin((\bar{u}+\vartheta)/2) + \sin(\bar{\Omega}/2) \cos(\bar{i}/2) \cos((\bar{u}+\vartheta)/2) \end{bmatrix} \quad (3)$$

Where, $\bar{\Omega}$ is the longitude of ascending node; $\bar{i}$ is the orbital inclination; $\bar{u}$ is the argument of perigee; and ϑ is the true anomaly.

The motion states of the tether can be described by the vector $\mathbf{r}_{ab}$ consisting of the attachment points a and b on the spacecraft, as shown in Fig. 1. As the tether is discretized into a multi-segmented elastic rod structure, the state quantities L, α and β of the tether in orbital coordinate system $C - x_c y_c z_c$ cannot be obtained directly from the Newton-Euler model, but can be obtained by calculating for $\hat{\mathbf{r}}_{ab}$ as explained in Appendix A. L denotes the length of tether. α and β denote the in-plane and out-of-plane angles of the tether concerning the objective plane, respectively. The attitude motion of the tether can take a quaternion form $\hat{\mathbf{q}}_{te}$ based on α and β as shown in (4).

$$\hat{\mathbf{q}}_{te} = \begin{bmatrix} \cos(\beta/2) \cos(\alpha/2) \\ -\sin(\beta/2) \sin(\alpha/2) \\ \sin(\beta/2) \cos(\alpha/2) \\ \cos(\beta/2) \sin(\alpha/2) \end{bmatrix} \quad (4)$$

Where, the quaternion $\hat{\mathbf{q}}_{te}$ is obtained from the orbital coordinate system $C - x_c y_c z_c$ to the tethered coordinate system $C - x_t y_t z_t$ by two primitive rotations with a 3–2 rotational sequence.

We use the Euler angle ψ, θ, ϕ to define the attitude of the body-fixed coordinate system relative to the tether coordinate system. The attitude motion model of tethered spacecraft is built in the body-fixed coordinate system using Euler's kinematic equations. To avoid the singularity problem of attitude modelling with the rotation matrix, we use quaternions to describe the rotation motion. The attitude unit quaternions are defined as $\hat{\mathbf{q}}_K \triangleq \{q_{K0}, \mathbf{q}_K\} \in \Re \times \Re^3, K = A, B$. The kinematic model of tethered spacecraft becomes

$$\dot{\mathbf{q}}_K = 0.5(\mathbf{q}_K^\times \boldsymbol{\omega}_K + q_{K0} \boldsymbol{\omega}_K) \quad (5a)$$

$$\dot{q}_{K0} = -0.5\mathbf{q}_K^T \boldsymbol{\omega}_K \quad (5b)$$

Where, $\boldsymbol{\omega}_K \in \Re^3$ represents the angular velocity of the body-fixed coordinate system $K - x_K y_K z_K$ with respect to the tether coordinate system $C - x_t y_t z_t$.

Then, the attitude dynamic model of tethered spacecraft is given by (5).

$$\mathbf{J}_K \dot{\boldsymbol{\omega}}_K = -\boldsymbol{\omega}_K^\times \mathbf{J}_K \boldsymbol{\omega}_K + \mathbf{r}_{Kk}^\times \mathbf{T}_K + \boldsymbol{\tau}_K + \mathbf{D}_K \quad (6)$$

Where, $\mathbf{r}_{Kk}^\times \mathbf{T}_K$ is the torque of tether tension on the attitude of tethered spacecraft. The tensions acting on the tethered spacecraft are $\mathbf{T}_A = \mathbf{T}_{a,1}$ and $\mathbf{T}_B = \mathbf{T}_{b,n}$. $\boldsymbol{\tau}_K$ and $\mathbf{D}_K$ are control torques and unknown bounded disturbance torques on spacecraft respectively.

To facilitate the design of attitude controllers, we define the attitude tracking error $\hat{\mathbf{e}}_K \triangleq \{e_{K0}, \mathbf{e}_K\} \in \Re \times \Re^3$. The rotation matrix $\mathbf{R}_K$ between the body-fixed coordinate system $K - x_K y_K z_K$ and the body-fixed reference coordinate system K_d takes the following form,

$$\mathbf{R}_K = (e_{K0}^2 - \mathbf{e}_K^T \mathbf{e}_K) \mathbf{I}_3 + 2\mathbf{e}_K \mathbf{e}_K^T - 2e_{K0} \mathbf{e}_K^\times \quad (7)$$

Based on the rotation matrix $\mathbf{R}_K$, the angular velocity of $K - x_K y_K z_K$ with respect to K_d define in (8),

$$\omega_{Ke} = \omega_K - \omega_d = \omega_K - \mathbf{R}_K \widehat{\omega}_d \quad (8)$$

Where, ω_d and $\widehat{\omega}_d$ are the reference angular velocity of K - $x_K y_K z_K$ and K_d , respectively. Then, we can calculate the kinematic and dynamic attitude error model of tethered spacecraft as

$$\dot{e}_K = 0.5(e_K^\times + e_{K0} \mathbf{I}_3) \omega_{Ke} \quad (9a)$$

$$\dot{e}_{K0} = -0.5 e_K^T \omega_{Ke} \quad (9b)$$

$$\begin{aligned} \mathbf{J}_K \dot{\omega}_{Ke} = & -\omega_K^\times \mathbf{J}_K \omega_K + \mathbf{J}_K (\omega_{Ke}^\times \mathbf{R}_K \omega_{Kd} - \mathbf{R}_K \dot{\omega}_{Kd}) \\ & + \mathbf{r}_{Kk}^\times \mathbf{T}_K + \boldsymbol{\tau}_K + \mathbf{D}_K \end{aligned} \quad (10)$$

In the previous modelling, the equations (1), (5), and (6) describe the dynamics of the tethered spacecraft system. The equations (9) and (10) describe the attitude tracking error model for tethered spacecraft.

2.2. Task objective

When the tether deploys and spins, it is required to use the thrust/tension control strategy along the centroid of tethered spacecraft. Many previous studies on the tethered spacecraft system have discussed the in-orbital plane motion of tether while ignoring the attitudes of the tethered spacecraft (Kang and Zhu (2019); Wang et al, 2019). However, considering the high coupling between the spacecraft attitudes and tether, the attitude control torques affect the tether motion.

To reduce the effect of attitudes on tether motion, we expect the attitudes of spacecraft concerning the tether system to be stable at zero, i.e., $\lim_{t \rightarrow \infty} \hat{\mathbf{q}}_K = [1, 0, 0, 0]^T$, $K = A, B$. During tether deployment and rotation, tethered spacecraft are subjected to the lowest tension torque on the x-axis. Hence, the x-axis of the body coordinate system is chosen as the underdrive axis. The other two actuated axes drive the 3-DOFs attitudes for stabilization.

3. Underactuated controller design

In the following, an underactuated tethered spacecraft attitude controller is proposed using the backstepping technique. The first step is to design an underactuated kinematic controller based on the attitude coupling relationship. And then, the dynamics tracking controller is designed to track the kinematic control input by backstepping. Finally, the asymptotic stability of the underactuated controller is demonstrated.

3.1. Kinematic controller design and stability analysis

In this case, the kinematic control input $\omega_{ea} = [\omega_{e2} \ \omega_{e3}]^T \in \mathbb{R}^2$ is designed for the attitude error model (9). We define a virtual error signal $\tilde{e}_a = e_d - e_a \in \mathbb{R}^2$, where e_d is the designed auxiliary converge signal, and actuated state signal is $e_a = [e_2 \ e_3]^T$.

Taking into account the differences in actuated characteristics, the kinematic error system (9) can be rewritten in the following form:

$$\dot{e}_u = 0.5(U + e_a^T \mathbf{E} \omega_{ea}) \quad (11a)$$

$$\dot{e}_a = 0.5(\mathbf{E} e_a \omega_{eu} + \mathbf{B} \omega_{ea}) \quad (11b)$$

where $e_u = e_1 \in \mathbb{R}^1$ is the underactuated error signal, $U = e_0 \omega_{eu} - e_2 \omega_{d3} + e_3 \omega_{d2}$, and $\mathbf{E}$, $\mathbf{B}$ are the rewritten matrixes of the following (12),

$$\mathbf{B} = \begin{bmatrix} e_0 & -e_u \\ e_u & e_0 \end{bmatrix}; \quad \mathbf{E} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad (12)$$

and $\omega_{eu} \in \mathbb{R}^1$ is the angular velocity of the underactuated axes.

Considering the novel kinematic error system (11), an underactuated kinematic control signal is proposed to stabilize the attitude tracking errors

$$\omega_{ea} = -k_a e_0 e_a + \kappa_1 \mathbf{E} e_d + \kappa_2 e_d \quad (13)$$

where the designed auxiliary signal e_d is defined by the oscillator-like relation as follow:

$$\dot{e}_d = \left(\frac{\dot{\xi}}{\xi} \right) e_d + \Lambda \mathbf{E} e_d, \quad e_d^T(0) e_d(0) = \xi^2(0) \quad (14)$$

and where the virtual variables $\kappa_1, \kappa_2, \Lambda$ and ξ are defined as $\kappa_1 = (k_u e_u + e_0 \omega_{e1}) / \xi^2, \kappa_2 = 2\dot{\xi} / e_0 \xi + k_a e_0, \Lambda = 0.5(k_a e_0 e_u + e_0 \kappa_1 + \omega_{eu})$, and $\xi = \gamma_0 \exp(-\gamma_1 t) + \gamma_2$. The controller parameters $k_u, k_a, \gamma_0, \gamma_1$ and γ_2 are positive constants in the virtual variables.

Remark 1. The first term in proposed controller (13) can guarantee the asymptotic convergence of actuated state. Meanwhile, the last two terms in Eq. (13) achieve the stability of the underactuated state error under oscillator-like constraints. The property of oscillator-like auxiliary signal e_d (14) can be given in the time derivative of $e_d^T e_d$:

$$\begin{aligned} d(e_d^T e_d) / dt &= 2e_d^T \dot{e}_d = 2e_d^T \left[\left(\frac{\dot{\xi}}{\xi} \right) e_d + \Lambda \mathbf{E} e_d \right] \\ &= 2 \left(\frac{\dot{\xi}}{\xi} \right) e_d^T e_d \end{aligned} \quad (15)$$

It can be easily proved that $e_d^T e_d = \xi^2$ by solving the differential (15) with the initial conditions $e_d^T(0) e_d(0) = \xi^2(0)$. Because of the coupling matrix $\mathbf{E}$, the second term in auxiliary signal (14) has been eliminated in time derivative of $e_d^T e_d$. Based on the above discussion, it can be concluded that $\|e_d\| = \xi$.

Remark 2. The exponential type function ξ is a constraint on the auxiliary signal e_d , which causes e_d to converge exponentially. ξ does not affect the control results of the kinematic attitude controller. However, a slower convergence time of ξ will result in a smaller attitude control input ω_{ea} . That is, we can choose smaller parameters of ξ to ensure that the control input is not too large.

Remark 3. The proposed kinematic controller (13) is differentiable and takes the form shown in (16). This differentiated form is used in the subsequent kinetic controller design to deal with angular acceleration information not being measurable.

$$\begin{aligned}\dot{\omega}_{ea} = & -k_a \dot{e}_a - k_a e_0 \dot{e}_a + (\dot{\kappa}_1 \mathbf{E} + \dot{\kappa}_2) \mathbf{e}_d \\ & + (\kappa_1 \mathbf{E} + \kappa_2) \dot{\mathbf{e}}_d\end{aligned}\quad (16)$$

where, the time derivatives of κ_1 and κ_2 are as follow:

$$\begin{aligned}\dot{\kappa}_1 = & (k_u \dot{e}_u + \dot{e}_0 \omega_{e1} + e_0 \dot{\omega}_{eu}) / \xi^2 - 2(k_u e_u + e_0 \omega_{eu}) \dot{\xi} / \xi^3 \\ \dot{\kappa}_2 = & 2(e_0 \xi \ddot{\xi} - \dot{e}_0 \xi \dot{\xi} - e_0 \dot{\xi}^2) / e_0^2 \xi^2 + k_a \dot{e}_0\end{aligned}\quad (17)$$

To obtain a closed-loop kinematic model on virtual errors, we substitute the proposed kinematic controller (13) for $\tilde{\mathbf{e}}_a$ into attitude error model (11) and (14).

$$\dot{e}_u = -\frac{1}{2} k_u e_u - \frac{1}{2} \tilde{\mathbf{e}}_a^T \mathbf{E} (\kappa_1 \mathbf{E} \mathbf{e}_d + \kappa_2 \mathbf{e}_d) \quad (18a)$$

$$\dot{\tilde{\mathbf{e}}}_a = \frac{1}{2} \left[-k_a e_0^2 \tilde{\mathbf{e}}_a + \mathbf{E} \tilde{\mathbf{e}}_a (\omega_{eu} + k_a e_0 e_u) + e_u \mathbf{E} (\kappa_1 \mathbf{E} \mathbf{e}_d + \kappa_2 \mathbf{e}_d) \right] \quad (18b)$$

Please see Appendix B for a detailed derivation of the closed-loop kinematic model (18). The following theorem holds regarding the stability of the closed-loop kinematic system.

Theorem 1. Considering the underactuated kinematic controller $\omega_{ea} \in \mathbb{R}^2$ (13) and closed-loop system model (18), if the designed auxiliary signal $\mathbf{e}_d$ is satisfied in (14), the equilibrium point $[0, 0, 0]^T$ of e_u and $\tilde{\mathbf{e}}_a$ is uniformly exponentially stable.

The proof of Theorem 1 has been given in Appendix C.

3.2. Dynamic controller design and stability analysis

After the kinematic underactuated control has been solved, it is necessary to design a dynamic controller for tethered spacecraft. The control objective for dynamics model (10) is to track the output ω_{Ke} of the tethered spacecraft dynamics system to the kinematic control input ω_{ea} , that is, to stabilize the dynamic tracking error variable $\boldsymbol{\eta} = \omega_{ea} - \omega_{Ke}$.

According to the definition of the dynamic tracking error variable $\boldsymbol{\eta}$, the transformed dynamic model can be given by the derivative of $\boldsymbol{\eta}$ in (19):

$$\mathbf{J} \dot{\boldsymbol{\eta}} = \mathbf{J} \dot{\omega}_{ea} + \mathbf{N}_a - \mathbf{M}_a - \boldsymbol{\tau} - \mathbf{D}_K \quad (19)$$

where $\mathbf{N}_a = \omega^\times \mathbf{J} \omega - \mathbf{J} (\omega_e^\times \mathbf{R} \omega_d - \mathbf{R} \dot{\omega}_d)$ and $\mathbf{M}_a = \mathbf{r}_{Kk}^\times \mathbf{T}_K$. The specific formula of $\dot{\omega}_{ea}$ can be found in Remark 3. $\mathbf{D}_K$ is the unknown bounded disturbance of spacecraft in space.

A natural logarithmic barrier function is incorporated into the controller gain to deal with the unknown distur-

bances, which can suppress disturbances in the absence of the estimating disturbance information. Compared with traditional disturbance estimation Ma et al. (2018), the revised barrier function achieves suppression by adaptively varying the controller gain in real-time. Simultaneously, it decreases the computational requirements, the complexity of the designed controller.

Definition 1. A continuous function H_B is called Positive-Definite Barrier Function (PDBF) if it satisfies the following hypotheses:

- (a) $x \in (-s, s) \rightarrow H_B(x) \in [0, \infty)$;
- (b) $H_B(0) = 0$; $\lim_{|x| \rightarrow \varepsilon} H_B = +\infty$; (c) H_B is a strictly monotonic increasing function on $[0, s)$, and a strictly monotonic decreasing function on $(s, 0]$.

The natural logarithmic PDBF is proposed as (20):

$$H_B(x) = \lambda \left(\ln \frac{s}{s - |x|} \right)^2 \quad (20)$$

$$\lambda > 0, s > 0$$

Remark 4. Different from the controller gain with traditional barrier function (Obeid et al. (2018)), the proposed PDBF is not only continuously differentiable in positive and negative domains, but also differentiable at zero. Moreover, the derivative of the novel PDBF near the zero neighborhood is much smaller. It guarantees that the controller will be not overly sensitive and not require excessive input compensation in the case of small disturbances.

Combining the aforementioned logarithmic barrier function, the underactuated control torque is designed for transformed dynamic model (19) based on the backstepping technique:

$$\begin{aligned}\boldsymbol{\tau} = & \mathbf{N}_a + \mathbf{J} \dot{\mathbf{u}}_d + k_p \mathbf{J} \boldsymbol{\eta} + 0.5 \mathbf{E} \mathbf{e}_a e_u + 0.5 \tilde{\mathbf{e}}_a \\ & + \mathbf{J} (\mathbf{K}(\boldsymbol{\eta}) \text{sign}(\boldsymbol{\eta}))\end{aligned}\quad (21)$$

with the logarithmic adaptive gain $\mathbf{K}(\boldsymbol{\eta})$

$$\mathbf{K}(\boldsymbol{\eta}) = \begin{cases} K_{Bi}(t), & \text{if } 0 < t < t_B \\ H_B(\eta_i), & \text{if } t > t_B \end{cases} \quad (22)$$

where, $\dot{K}_{Bi}(t) = k_{BF} |\eta_i|$, $i = 1, 2, k_{BF} > 0$, t_B is the smallest time of $-s/2 \leq \eta_i \leq s/2$.

To facilitate the proof of the closed-loop system stability, we substitute the auxiliary relations $\omega_{Ke} = \omega_{ea} - \boldsymbol{\eta}$ into the open-loop kinematic and dynamical models (11) and (19) to obtain the closed-loop error model. After the same process as for the kinematic system, the closed-loop error model with e_u and $\tilde{\mathbf{e}}_a$ can be expressed as follows:

$$\dot{e}_u = -\frac{1}{2} k_u e_u - \frac{1}{2} \tilde{\mathbf{e}}_a^T \mathbf{E} (\kappa_1 \mathbf{E} \mathbf{e}_d + \kappa_2 \mathbf{e}_d) + \frac{1}{2} \boldsymbol{\eta}^T \mathbf{E} \mathbf{e}_a \quad (23a)$$

$$\dot{\tilde{\mathbf{e}}}_a = \frac{1}{2} \left[-k_a e_0^2 \tilde{\mathbf{e}}_a + \mathbf{E} \tilde{\mathbf{e}}_a (\omega_{eu} + k_a e_0 e_u) + e_u \mathbf{E} (\kappa_1 \mathbf{E} \mathbf{e}_d + \kappa_2 \mathbf{e}_d) + \boldsymbol{\eta} \right] \quad (23b)$$

Substituting the control torque input (21) into the transformation dynamics model (23), we can obtain a closed-loop error model with respect to η :

$$\mathbf{J}\dot{\eta} = -k_{\eta}\mathbf{J}\eta - 0.5\mathbf{E}e_a e_u - 0.5\tilde{e}_a - \mathbf{J}(\mathbf{K}(\eta)\text{sign}(\eta)) - \mathbf{d}_K \quad (24)$$

where, $\mathbf{d}_K = \mathbf{M}_a + \mathbf{D}_K$. The tension torque $\mathbf{M}_a$ is considered as part of the disturbance of tethered spacecraft for the controller design. This hypothesis solves the problem of tension torque being difficult to measure in space. The detailed derivations of the closed-loop error model (23) and (24) have been obtained by a similar process to the derivations of the closed-loop kinematic model (18) presented in Appendix B.

Theorem 2. Considering the closed-loop error system with $e_u, \tilde{e}_a$ and η in (23) and (24), if the underactuated control torque input τ is provided in (21), and the designed auxiliary signal e_d is satisfied in (14), the closed-loop error states are uniformly asymptotic stable.

The proof of Theorem 2 has been given in Appendix D. To facilitate the understanding of the proposed underactuated control scheme, the schematic diagram of the closed-loop system is shown in Fig. 2.

4. Cases studies and simulations

In this section, to verify the efficiency of the underactuated controller in the field of tethered spacecraft, we have used the proposed controller to simulate the deployment and spin-up process. To reduce the coupling effect of attitude control forces to the translational motion of the teth-

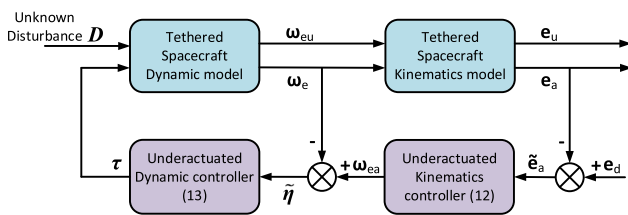


Fig. 2. Schematic diagram of the closed-loop system.

Table 1
Parameters of tethered spacecraft.

Parameters	Values
Orbital height $\ r_C\ $	300km
Spacecraft masses m_K	200kg
Inertia moment of spacecraft $\mathbf{J}_K$	$\text{diag}([300, 300, 300])$
Tether length L	1000m
Tether elements masses m_i	5kg
Damping coefficient of tether	0.3
Axial rigidity of tether EA	$8.3 \times 10^5 Pa$
Number of tether element points n	3
Initial attitude quaternion $\hat{q}_{K0}$	[0.911; -0.2; 0.3; 0.2]
Vectors of r_{Aa}, r_{Bb}	$r_{Aa} = -r_{Bb} = [2; 0; 0]$

ered spacecraft, we simplify the installation of attitude nozzles using the proposed underactuated control strategy in Fig. 3. The red and green arrows indicate the actuated angular velocities and torque inputs in Fig. 3. Moreover, the yellow arrow indicates the underdrive angular velocity ω_x . The unknown bounded disturbance is chosen as $\mathbf{D}_{Ki} = 10^{-2} \times [-0.15 + 0.2 \cos(0.06t) - 0.15 \sin(0.2t)]$, $i = 1, 2, 3$. The parameters of tethered spacecraft system are shown in Table 1.

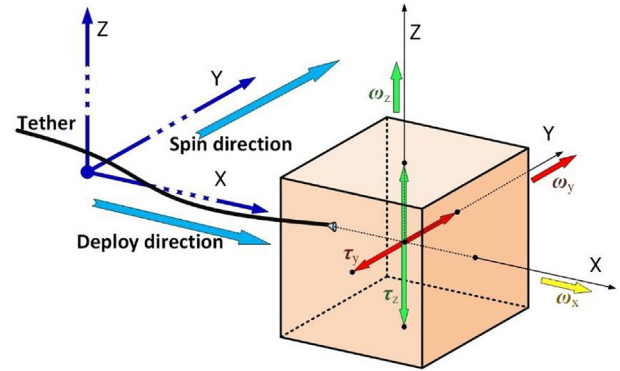


Fig. 3. Schematic of attitude control torques layout.

Table 2
Parameters of controllers.

Parameters	Values
Tension controller parameter k	3.3×10^{-4}
Tension controller parameter k_p	6×10^{-4}
Tension controller parameter k_d	0.35
Kinematic controller parameter k_u	8.5
Kinematic controller parameter k_a	0.6
Dynamic controller parameter k_{η}	300
Auxiliary converge signal parameter γ_0	0.04
Auxiliary converge signal parameter γ_1	0.2
Auxiliary converge signal parameter γ_2	0.001
Barrier function parameter s	0.04
Adaptive gain parameter k_{BF}	0.1

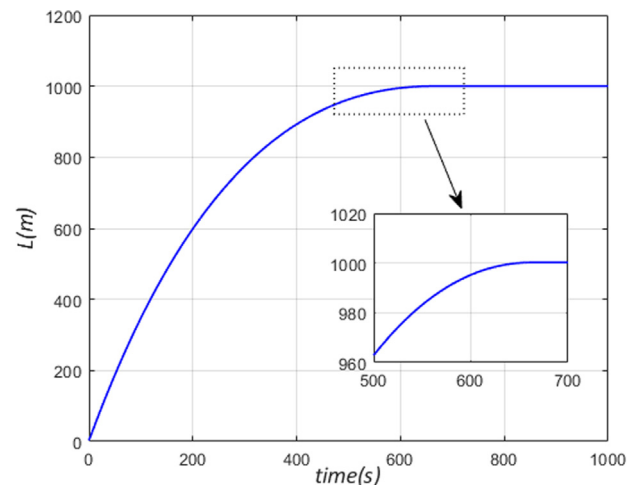


Fig. 4. Tether length, L

4.1. Case 1: Deployment

Inspired by (Kang and Zhu (2019)), we apply the proposed linear PD tension controller with gravity compensation to the tethered spacecraft dynamics model established in this paper to simulate the tether deployment. The specific form of the linear PD controller with gravity compensation in deployment is as follows:

$$T = kL + k_p(L - L_K) + k_d\dot{L} \quad (25)$$

Where, L_K is the desired tether length. k , k_p , k_d are the parameters of the tension controller. Unlike the tethered model in (Kang and Zhu (2019)) (spacecraft as a lumped mass), we consider the 3DOFs attitude in our modeling and simulation work. The proposed underactuated attitude control strategy is applied to the deployment with (25) to verify the effectiveness of the underactuated attitude controller. Simultaneously, the impact of the underactuated attitude controller on deployment is also illustrated. The controller parameters are set in the Table 2:

The simulation results of the deployment process are discussed in this case. It can be seen in Fig. 4 that the tension controller effectively deploys the tether to the desired length of 1000 m. Because of the quick beginning speed, the tether is fully extended in roughly 650 s.

Fig. 5 show the attitude quaternion of tethered spacecraft under the proposed underactuated controller. As can be seen in Figs. 5 and 6, the underactuated state q_1 is stabilized by the joint control of the actuated terms ω_y and ω_z . The cost of stability of q_1 is the presence of small oscillations in the actuated states q_2 and q_3 before convergence to the stable domain, which is caused by the oscillator-like relationship in (14). The oscillation amplitude of q_2 and q_3 is constrained by the exponential convergence function ξ . The oscillation frequency is affected by the parameters k_u and k_a of the oscillator-like relationship. The controller parameters k_u and k_a affect the convergence speed of the underactuated and actuated states, respectively. Because the q_1 is converged by the exponential constraint oscillations of q_2 and q_3 , the stable control of q_1 should be completed before the constraint ξ exponentially converges to the zero neighborhood. When adjusting the controller parameter, the larger k_u is, the faster convergence of the underactuated state is, the higher oscillation frequencies of q_2 and q_3 are, that like the proportional control law. We choose the k_u which can make the convergence time of the underactuated state q_1 approximate to the convergence time of the exponential constraint ξ . In this way, the stable convergence of q_1 can be completed in time with the minimum oscillation frequency, as shown in Fig. 5. As q converges to zero, q_0 gradually converges to 1, so as to satisfy the constraint of quaternion norm.

Next, the time histories of angular velocities and control torques are shown in Figs. 6 and 7. Consistent with the preceding underactuated control strategy, attitude control during deployment is achieved by driving the angular

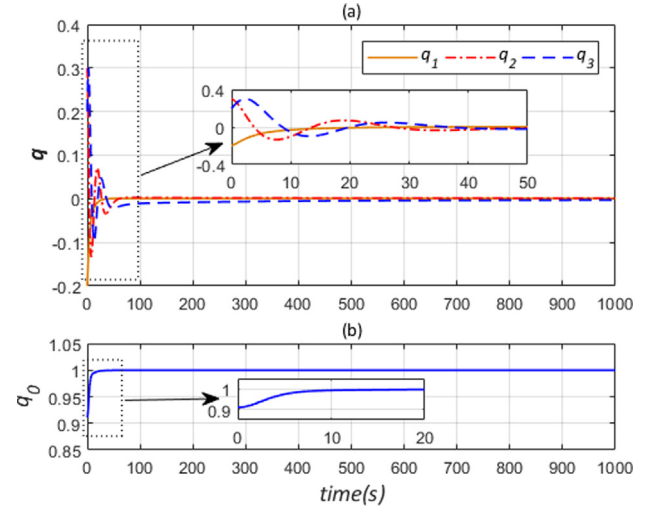


Fig. 5. Attitude quaternion, $q = [q_0; q]$, (a) q ; (b) q_0 .

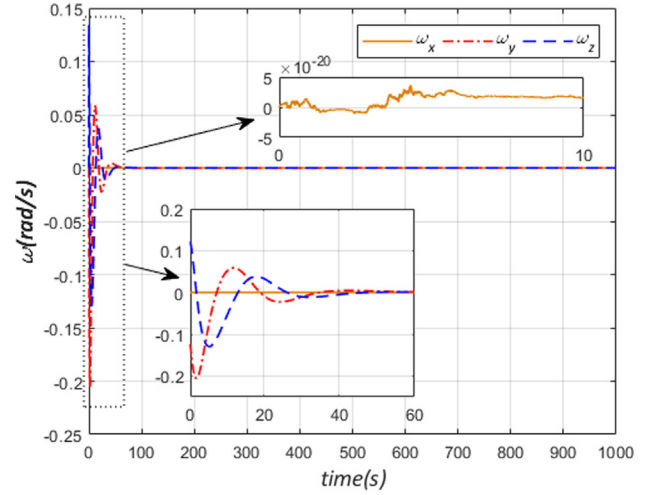


Fig. 6. Angular velocity of body-fixed axes, ω

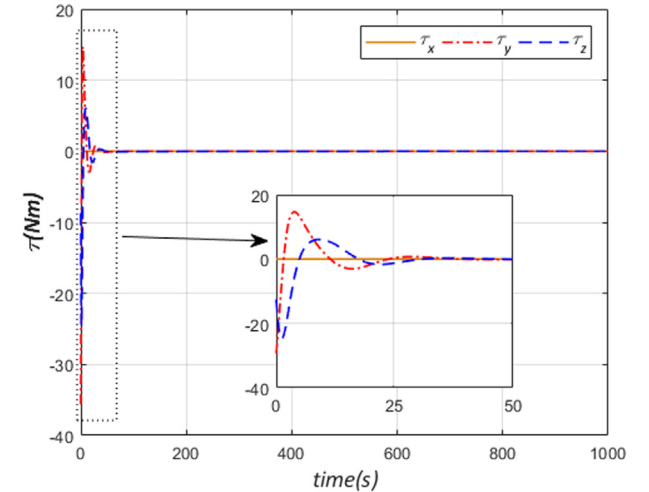


Fig. 7. Underactuated control torque input, τ

velocities ω_y and ω_z to no more than ± 0.2 rad/s. The angular velocity ω_x of the underactuated axis has a tiny oscillation when q_1 converges, and then it stabilizes in the zero neighborhood with a negligible amplitude magnitude of no more than 10^{-20} . The attitude control torques τ_y and τ_z do not exceed $35N \cdot m$, and τ_x is always $0N \cdot m$.

The closed-loop dynamic error η is plotted in Fig. 8. As with the attitude quaternion $\mathbf{q}$, the η converges to a zero neighborhood within 60 s. In this case, the closed-loop dynamic error η satisfies the barrier function constraint $-s_2/2 \leq \eta_i \leq s_1/2$, so that the adaptive control gain is always barrier function-type as shown in Fig. 9. Since the adaptive gain $\mathbf{K}$ is a barrier function with $\eta = [\eta_y; \eta_z]$, $\mathbf{K}$ is positively correlated with η . Meanwhile, the trend of the convergence curve of $\mathbf{K}$ are the same as trend of η . When η converges to zero, the barrier function-type gain also stable to zero.

Combined with the above discussion of simulation results, it can be seen that the proposed underactuated tethered attitude controller achieves stable convergence of $\mathbf{q}$ quickly and efficiently during the deployment. Meanwhile, the tether has been deployed smoothly to the desired tether length (1000m). As a result, there was no tether spring back or incomplete deployment.

4.2. Case 2: Spin-up

In Case 2, numerical simulations of the tethered system are performed during spin-up along the orbital direction to validate the effectiveness of the proposed controller. Because of the rotation matrix theory, the classical model of tethered systems suffers from singularity problems when the out-of-plane angle $\beta = \pi/2$. Thus, most published studies have focused on the rotational motion in the orbital plane (neglecting the out-of-orbital plane angle β) (Gou et al., (2018)).

To circumvent the singularity problem at specified angles, the quaternion theory was applied for coordinate frame rotation. As a result, out-of-orbital plane motion is no longer ignored, and spin-up along the orbital direction is researched for the first time. The rotational plane is perpendicular to the orbital direction in Case 2.

The initial state of the tether system is assumed to be fully deployed 1000m along the geographic vertical direction. The parameters of the tethered spacecraft system are defined as in Case 1, see Table 1. A spinning tangential constant thrust controller is utilized for the spin-up along the orbital direction, inspired in (Gou et al., (2018)). The open-loop spin-up controller is shown in the following (26):

$$F_z = \begin{cases} f_z, & \text{if } \dot{\beta} < \varpi_d \\ 0, & \text{other} \end{cases} \quad (26)$$

where, the spin-up thrust $f_z = 180N$, ϖ_d is the desired angular velocity. The spin-up controller causes the tethered to spin at a classically uniform acceleration. The param-

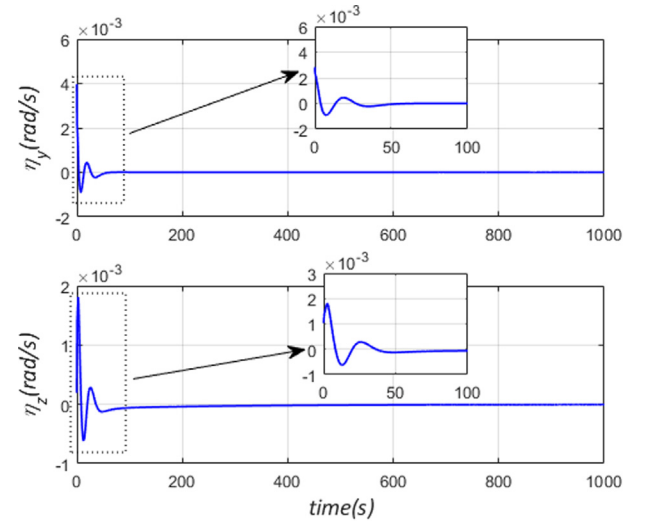


Fig. 8. Closed-loop dynamic error, $\eta = [\eta_y; \eta_z]$

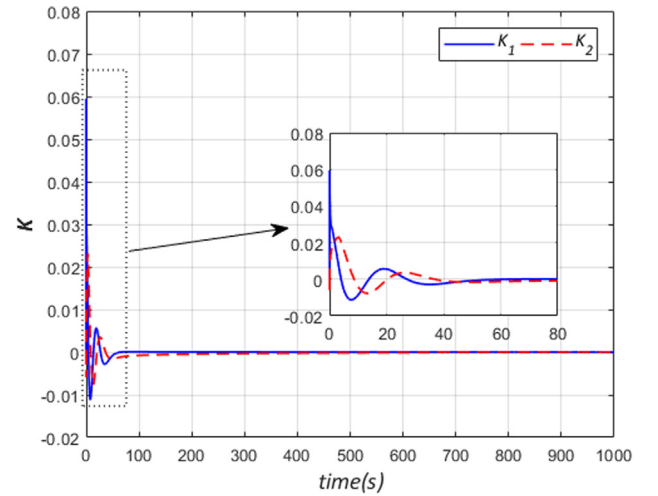


Fig. 9. Adaptive gain, $\mathbf{K}$

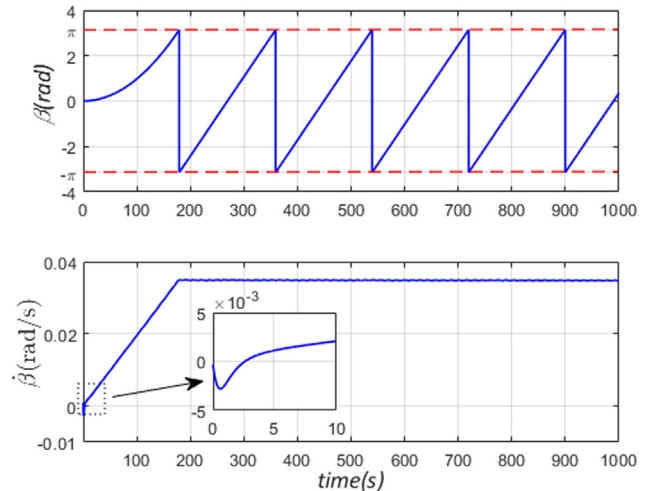


Fig. 10. Out-of-plane angle and angular velocity during spin-up.

ters of the attitude controller are the same as those in Case 1, see Table 2.

The simulation results of spin-up are given as follow. Fig. 10 represents the time history of the out-of-plane angle β during spin-up. The tether is accelerated uniformly to the desired angular speed (around 0.035 rad/s) in about 180 s. Because the out-of-plane angle β is calculated from the tether attachment point a and b , the attitude stability control at the beginning of the spin-up affects the position of a and b , which appears as a slight tick in the curve of the out-of-angular velocity.

Figs. 11–13 depicts the time histories of attitude quaternion, angular velocity and control torque, respectively. The subplot of Fig. 11 shows that the actuated states q_2 and q_3 converge gradually to the zero neighborhood. Since the controller parameters of both cases are the same, the convergence time of attitude quaternion is also about 60 s. Considering the extremely small tension torque in x-axis,

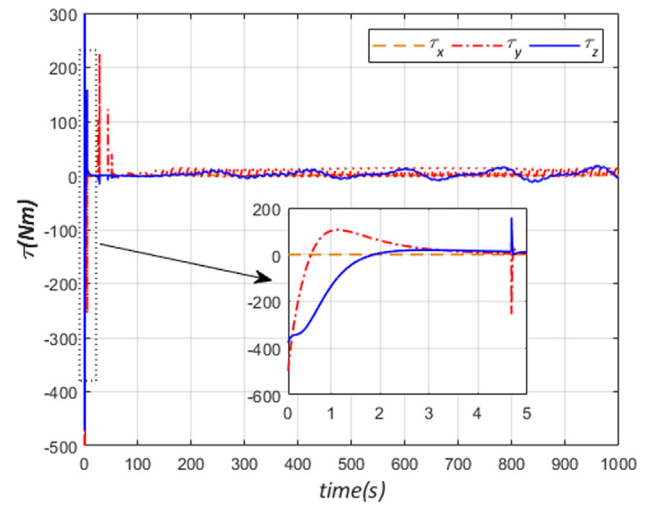


Fig. 13. Underactuated control torque input, τ

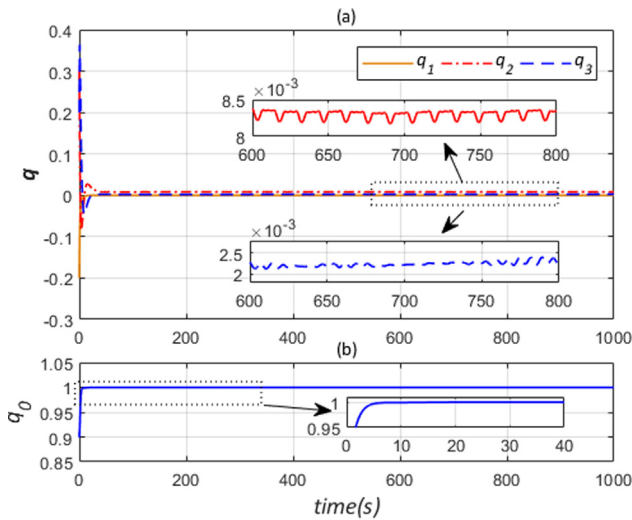


Fig. 11. Attitude quaternion, $\mathbf{q} = [q_0; \mathbf{q}]$, (a) $\mathbf{q}$; (b) q_0 .

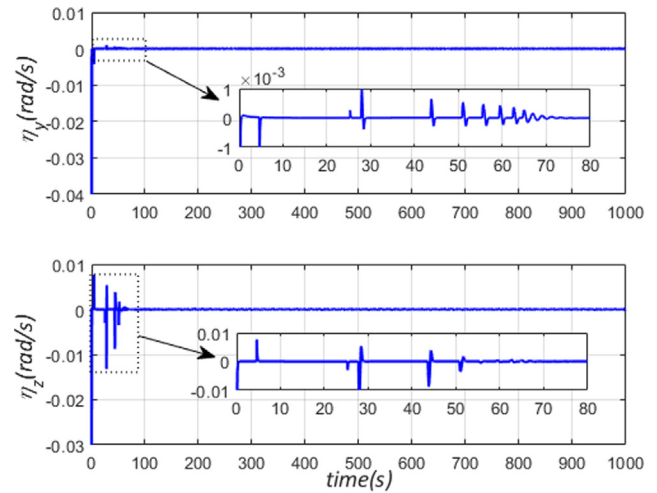


Fig. 14. Closed-loop dynamic error, $\boldsymbol{\eta} = [\eta_y; \eta_z]$

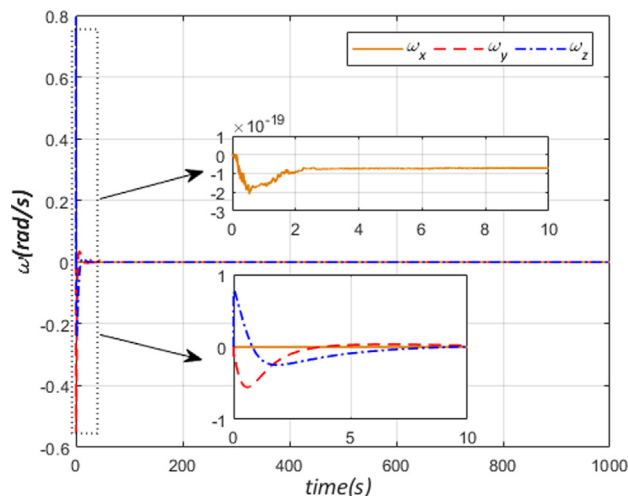


Fig. 12. Angular velocity of body-fixed axes, $\boldsymbol{\omega}$

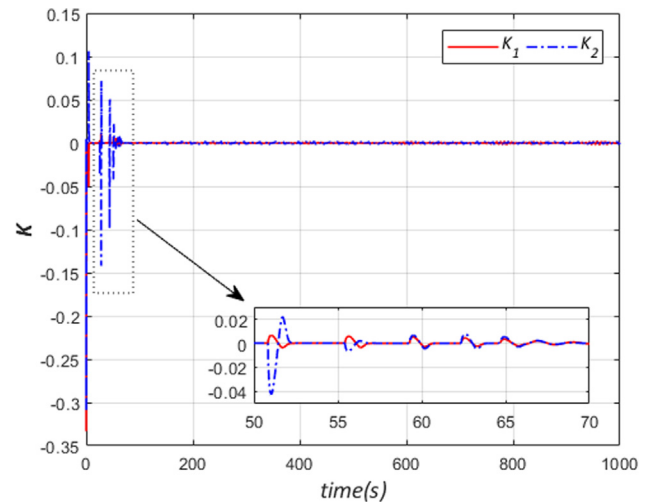


Fig. 15. Adaptive gain, $\mathbf{K}$ of underactuated controller.

q_1 converges to 0 without too many oscillations. However, there are small oscillations of q_2 and q_3 due to the tension torque as red and blue lines in Fig. 11.

In Fig. 12, spacecraft attitude during spinning is controlled by driving the angular velocities ω_y and ω_z to no more than ± 0.6 rad/s. The tiny oscillation of ω_x at the beginning of spin-up can be ignored with no more than 10^{-19} . In Fig. 13, the shakes of control torque τ_y and τ_z at the spin acceleration are compensating for the abrupt tension torque, which does not exceed 500 Nm in the whole control process. When we design the adaptive anti-disturbance term, the tether tension torque is regarded as unknown external disturbance. Therefore, the barrier function-type gain restrains the tension torque of the tethered spacecraft. In the early stage of the tether spin-up, the tether gradually tenses and the tension increases. With the swing of spacecraft attitude, the tension torque will have continuous shakes. The continuous shakes in the first 100 s in Fig. 13 are the small oscillation of the closed-loop error formed by the tension torque under the inhibition of the barrier function. Since the x-axis is underactuated, τ_x is always 0 Nm as shown by the orange dotted line.

The time history of the closed-loop dynamic error η is given in Fig. 14. η converges to within the zero neighborhood at the end of spin-up. Figs. 14 and 15 show η and adaptive gain K have some peaks at the beginning of the

spin-up (within 80 s). These peaks are caused by sudden changes in tension torque when tether tautness. When designing the dynamics controller, we consider the tension torque as an external disturbance. As η converges to 0.02 for the first time, the barrier function starts to affect the adaptive gain. By adapting the controller gain, the dynamic states η is guaranteed not to exceed the constraint of $-s_1/2 \leq \eta_i \leq s_2/2$ under subsequent external disturbances. The peaks in Fig. 15 corresponds to the peaks of η . It can be seen that the proposed barrier function adaptive law can effectively suppress the influence brought by external disturbance of tethered spacecraft.

To show the spinning process more clearly, a three-dimensional simulation result of the tether spinning along the orbital direction is given in Fig. 16. Red dots 1, 2 and 3 indicate the initial positions of the tether discrete mass points. A and B indicate the initial positions of the two sub-spacecraft. The blue to yellow gradient curves indicate the trajectories of five targets. The four folded lines in blue, light blue, green and yellow show the change in rotation position of the tether with time in turn. The tethered system always rotates in the direction of the orbital motion with the lateral deviation not exceeding 10 m. In conclusion, the attitude is effectively controlled, and the attitude control torque does not affect the spin-up motion.

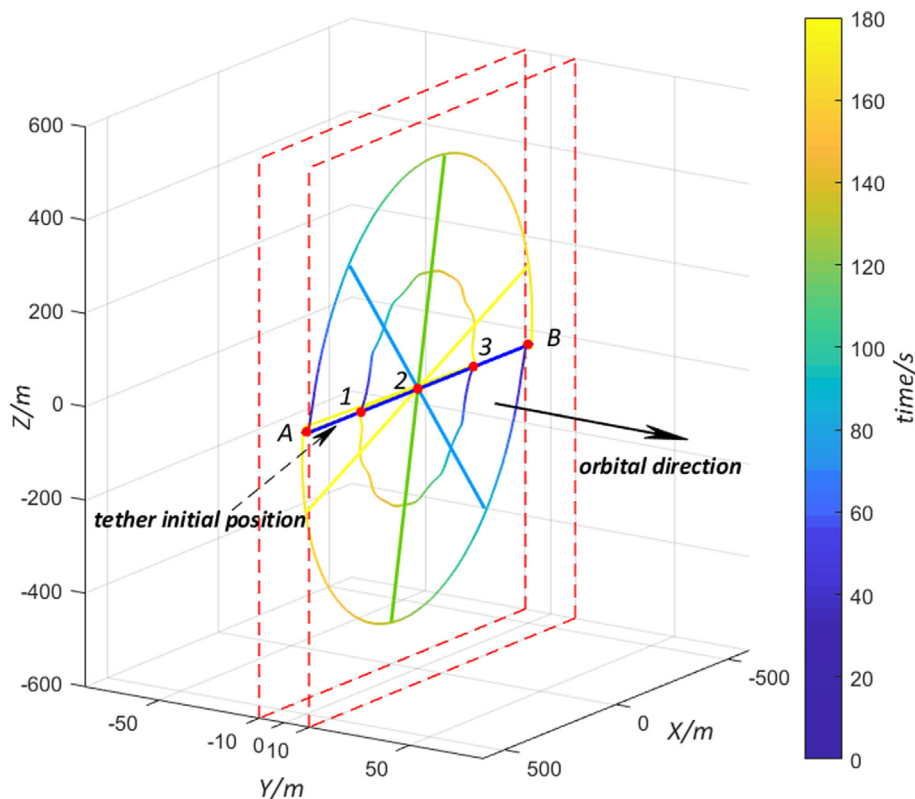


Fig. 16. Three-dimensional diagram of spin-up along orbital direction.

5. Conclusion

The article develops the dynamics and control for the tethered system stabilization with the coupling of tethered spacecraft attitude to the flexible tether during deployment and spin-up. The mathematical model of the tethered system is built by the Newton-Euler method with quaternion transformations, which avoids the singularity problems of out-of-plane motion and attitude. An underactuated attitude controller consisting of the kinematics controller designed on attitude coupling and the dynamics controller based on backstepping theory is presented to simplify the effect of attitude control on tethered movement. Among these, the controller gain based on a natural logarithmic barrier function is proposed to suppress unknown bounded external disturbances with no need for the estimated disturbance signal. Finally, the stability analysis and numerical simulations demonstrate the robustness and effectiveness of the proposed underactuated closed-loop control strategy. It is also indicated that the underactuated control strategy has a negligible influence on tether deployment and spinning motion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. According to the preceding discussion, the vector $\widehat{\mathbf{r}}_{ab}$ is solved to obtain the tethered states in the orbital coordinate system. Based on the structure of the tethered spacecraft system shown in Fig. 1, $\widehat{\mathbf{r}}_{ab}$ can be calculated using the following equation:

$$\widehat{\mathbf{r}}_{ab} = \widehat{\mathbf{r}}_B + \widehat{\mathbf{r}}_{Bb} - \widehat{\mathbf{r}}_A - \widehat{\mathbf{r}}_{Aa} \quad (\text{A1})$$

where, $\widehat{\mathbf{r}}_A$, $\widehat{\mathbf{r}}_B$ are the radius vectors of tethered spacecraft, and $\widehat{\mathbf{r}}_{Aa}$, $\widehat{\mathbf{r}}_{Bb}$ are the vectors from the centroids of spacecraft to the tethered connection points. The equation (A1) is described in the orbital coordinate system. Using quaternion multiplication to calculate the coordinate transformation, we can obtain the following relationship:

$$\widehat{\mathbf{r}}_K = \widehat{\mathbf{q}}_{or}^* \circ \mathbf{r}_K \circ \widehat{\mathbf{q}}_{or}; \quad \widehat{\mathbf{r}}_{Kk} = \widehat{\mathbf{q}}_{te} \circ (\widehat{\mathbf{q}}_K \circ \widetilde{\mathbf{r}}_{Kk} \circ \widehat{\mathbf{q}}_K^*) \circ \widehat{\mathbf{q}}_{te}^* \quad (\text{A2})$$

where $K = A, B, k = a, b$. And, $\widehat{\mathbf{q}}_{or}$, $\widehat{\mathbf{q}}_{te}$, $\widehat{\mathbf{q}}_K$ are the orbital quaternion, tether quaternion, and attitude quaternion respectively, described by (3), (4), and (5).

After that, the tether length L is calculated as the 2-norm of $\widehat{\mathbf{r}}_{ab}$:

$$L = \|\widehat{\mathbf{r}}_{ab}\| \quad (\text{A3})$$

The orbital in-plane angle α and out-plane angle β are each calculated by the inverse tangent as follows.

$$\alpha = \arctan 2(y_{ab}/x_{ab}); \beta = \arctan 2(z_{ab}/(x_{ab}^2 + y_{ab}^2)^{1/2}) \quad (\text{A4})$$

Appendix B. Closed-loop kinematic system development

Considering the underactuated state in rewritten kinematic error system (11), the controller (13) is substituted

$$\dot{e}_u = 0.5(e_0\omega_{eu} - e_2\omega_{d3} + e_3\omega_{d2} - k_a e_0 e_a^T \mathbf{E} e_a + e_a^T \mathbf{E}(\kappa_1 \mathbf{E} e_d + \kappa_2 e_d)) \quad (\text{B1})$$

Utilizing the fact that $e_a^T \mathbf{E} e_a = 0$ and $\omega_{d2} = \omega_{d3} = 0$, we can simplify the (B1) as follows:

$$\dot{e}_u = 0.5(e_0\omega_{eu} + e_a^T \mathbf{E}(\kappa_1 \mathbf{E} e_d + \kappa_2 e_d)) \quad (\text{B2})$$

Substituting $e_a = e_d - \widetilde{e}_a$ into (B2) with the fact that $\mathbf{E}\mathbf{E} = -\mathbf{I}_2$ and $\|e_d\| = \xi$, the closed-loop kinematic error for $\dot{e}_u$ is obtained as follows:

$$\dot{e}_u = 0.5(e_0\omega_{eu} - \|e_d\|^2[(k_u e_u + e_0\omega_{eu})/\xi^2] - \widetilde{e}_a^T \mathbf{E}(\kappa_1 \mathbf{E} e_d + \kappa_2 e_d)) \quad (\text{B3})$$

$$\dot{e}_u = 0.5(k_u e_u - \widetilde{e}_a^T \mathbf{E}(\kappa_1 \mathbf{E} e_d + \kappa_2 e_d)) \quad (\text{B4})$$

Then, taking the time derivative of $\widetilde{e}_a = e_d - e_a$, we substitute (14) and (11b) for $\dot{e}_d$ and $\dot{e}_a$, respectively.

$$\dot{\widetilde{e}}_a = \left(\dot{\xi}/\xi\right)e_d + \Lambda \mathbf{E} e_d - 0.5(\mathbf{E} e_a \omega_{eu} + \mathbf{B} \omega_{ea}) \quad (\text{B5})$$

Substituting $\Lambda = 0.5(k_a e_0 e_u + e_0 \kappa_1 + \omega_{eu})$, we can rewrite (B5) as follow:

$$\begin{aligned} \dot{\widetilde{e}}_a &= \left(\dot{\xi}/\xi\right)e_d + 0.5(k_a e_0 e_u + e_0 \kappa_1 + \omega_{eu})\mathbf{E} e_d - 0.5(\mathbf{E} e_a \omega_{eu} + \mathbf{B} \omega_{ea}) \\ &= \left(\dot{\xi}/\xi\right)e_d + 0.5(k_a e_0 e_u + e_0 \kappa_1 + \omega_{eu})\mathbf{E} e_d - 0.5\omega_{eu}\mathbf{E} e_a \\ &\quad + 0.5k_a e_0 \mathbf{B} e_a - 0.5\kappa_1 \mathbf{B} \mathbf{E} e_d - 0.5\kappa_2 \mathbf{B} e_d \\ &= \left(\dot{\xi}/\xi\right)e_d + 0.5(k_a e_0 e_u + e_0 \kappa_1)\mathbf{E} e_d + 0.5\omega_{eu}\mathbf{E} \widetilde{e}_a \\ &\quad + 0.5k_a e_0^2 e_a - 0.5e_0 \kappa_1 \mathbf{E} e_d - 0.5e_u \kappa_1 e_d - 0.5\left(2\dot{\xi}/e_0 \xi + k_a e_0\right)\mathbf{B} e_d \end{aligned} \quad (\text{B6})$$

$$\begin{aligned} \dot{\widetilde{e}}_a &= 0.5k_a e_0 e_u \mathbf{E} \widetilde{e}_a + 0.5\omega_{eu}\mathbf{E} \widetilde{e}_a - 0.5k_a e_0^2 \widetilde{e}_a \\ &\quad - 0.5e_u \kappa_1 e_d + e_u \dot{\xi} \mathbf{E} e_d / e_0 \xi + 0.5k_a e_0 e_u \mathbf{E} e_d \end{aligned} \quad (\text{B7})$$

where, $\mathbf{B} = e_0 \mathbf{I}_2 - e_u \mathbf{E}$ is utilized to simplify the closed-loop system. The last three terms in the closed-loop system of $\dot{\widetilde{e}}_a$ (B7) can be combined by taking out $0.5e_u \mathbf{E}$:

$$\begin{aligned} \dot{\widetilde{e}}_a &= -0.5k_a e_0^2 \widetilde{e}_a + 0.5(k_a e_0 e_u + \omega_{eu})\mathbf{E} \widetilde{e}_a \\ &\quad + 0.5e_u \mathbf{E}(\kappa_1 \mathbf{E} e_d + \kappa_2 e_d) \end{aligned} \quad (\text{B8})$$

Relying on the preceding discussion, the equations (B4) and (B8) describe the closed-loop kinematic error system.

Appendix C. Proof of theorem 1

PROOF: To illustrate the convergence of the closed-loop kinematic states, we choose the positive definite Lyapunov function candidate as follows:

$$V_1 = e_u^2 + \tilde{e}_a^T \tilde{e}_a \quad (C1)$$

Taking the derivative of Lyapunov function V_1 :

$$\dot{V}_1 = 2e_u \dot{e}_u + 2\tilde{e}_a^T \dot{\tilde{e}}_a \quad (C2)$$

Substituting the closed-loop kinematic model (18) into Eq. (C2), we have

$$\begin{aligned} \dot{V}_1 &= e_u \left[-k_u e_u - \tilde{e}_a^T E (\kappa_1 E e_d + \kappa_2 e_d) \right] \\ &+ \tilde{e}_a^T \left[-k_a e_0^2 \tilde{e}_a + E \tilde{e}_a (\omega_{eu} + k_a e_0 e_u) + e_u E (\kappa_1 E e_d + \kappa_2 e_d) \right] \\ &= -k_u e_u^2 - k_a e_0^2 \tilde{e}_a^T \tilde{e}_a + \tilde{e}_a^T E \tilde{e}_a (\omega_{eu} + k_a e_0 e_u) \leq -\varepsilon V_1 \end{aligned} \quad (C3)$$

where, $\varepsilon = \min \{k_u, k_a e_0^2\}$. It can be seen that when $k_u, k_a > 0$, $\dot{V}_1 \leq -\varepsilon V_1 < 0$ holds and the closed-loop kinematic system is uniformly exponentially stable.

Appendix D. Proof of theorem 2

PROOF: A Lyapunov function candidate is chosen as:

$$\begin{aligned} V_2 &= \frac{1}{2} \eta^T J \eta \\ &+ \frac{1}{2} (H_B(\eta) - H_B(0))^T J (H_B(\eta) - H_B(0)) \end{aligned} \quad (D1)$$

Taking the derivative of the Lyapunov function $V = V_1 + V_2$:

$$\begin{aligned} \dot{V} &= 2e_u \dot{e}_u + 2\tilde{e}_a^T \dot{\tilde{e}}_a + \eta^T J \dot{\eta} \\ &+ [(H_B(\eta) - H_B(0))^T J \dot{H}_B(\eta)] \end{aligned} \quad (D2)$$

Substituting the closed-loop error model (23) and (24) into (D2), we have

$$\begin{aligned} \dot{V} &= e_u \left[-k_u e_u - \tilde{e}_a^T E (\kappa_1 E e_d + \kappa_2 e_d) + \eta^T E e_a \right] \\ &+ \tilde{e}_a^T \left[-k_a e_0^2 \tilde{e}_a + E \tilde{e}_a (\omega_{eu} + k_a e_0 e_u) + e_u E (\kappa_1 E e_d + \kappa_2 e_d) + \eta \right] \\ &+ \eta^T [J \dot{u}_d + N_a - M_a - \tau_K - d_K] + [(H_B(\eta) - H_B(0))^T J \dot{H}_B(\eta)] \\ &< -k_u e_u^2 - k_a e_0^2 \tilde{e}_a^T \tilde{e}_a - k_p \eta^T J \eta + |\eta^T [d_{K \max} - \eta^T J [H_B(\eta) \text{sign}(\eta)] \\ &+ H_B(\eta)^T J \dot{H}_B(\eta)]| \end{aligned} \quad (D3)$$

Since $H_B(\eta)$ is the logarithmic barrier function, (D3) can be derived by the underactuated controller (21):

$$\begin{aligned} \dot{V} &\leq -k_u e_u^2 - k_a e_0^2 \tilde{e}_a^T \tilde{e}_a - k_p \eta^T J \eta + |\eta^T [d_{K \max} - \eta^T J [H_B(\eta) \text{sign}(\eta)] + H_B(\eta)^T J \dot{H}_B(\eta)]| \\ &\leq -k_u e_u^2 - k_a e_0^2 \tilde{e}_a^T \tilde{e}_a - [d_{K \max} + J H_B(\eta) + k_p J \eta]^T |\eta| \\ &\quad - [-d_{K \max} + J H_B(\eta) + k_p J \eta]^T \text{diag} \left(\frac{2\lambda_{s_i} \ln [s_i / (s_i - |\eta_i|)]}{(s_i - |\eta_i|)^2} \right) H_B(\eta) \end{aligned} \quad (D4)$$

After combining and simplifying the Eq. (D4), we get the following inequality:

$$\begin{aligned} \dot{V} &\leq -k_u e_u^2 - k_a e_0^2 \tilde{e}_a^T \tilde{e}_a - \zeta_s^T |\eta| - \zeta_s^T \zeta H_B(\eta) \\ &\leq -\varepsilon V_1 - \sqrt{2} \min(1, \zeta) \zeta_s^T \left(\frac{|\eta|}{\sqrt{2}} + \frac{H_B(\eta)}{\sqrt{2}} \right) \leq -\varepsilon V_1 - \zeta_{\min} V_2^{\frac{1}{2}} \end{aligned} \quad (D5)$$

where,

$$\begin{aligned} \zeta_s &= -d_{K \max} + J K_B(\eta) + k_p J \eta, \zeta = \text{diag} \left(\frac{2\lambda_{s_i} \ln [s_i / (s_i - |\eta_i|)]}{(s_i - |\eta_i|)^2} \right), \zeta_{\min} \\ &= \min [\sqrt{2} \min(1, \zeta) \zeta_s^T]_{1,2}. \end{aligned}$$

It can be obtained that ζ_s, ζ and $\zeta_{\min}$ are the positive constants. With the underactuated controller (21), $\dot{V} < 0$ holds. The state of closed-loop error system can be uniformly asymptotic stable in the presence of unknown disturbance by the Lyapunov theorem.

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